Cevahir Koprulu

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GitHub | LinkedIn | Expected graduation: May 2027

RESEARCH PROFILE

My research focuses on improving the sample efficiency of reinforcement learning through automated curriculum generation and reward synthesis. My research experience in industry spans foundation-model post-training, preference learning, multi-agent self-play, and teacher-student knowledge distillation for RL agents.

EDUCATION

The University of Texas at Austin Aug 2021 - Present
PhD, Electrical and Computer Engineering, expected May 2027
Advisor: Ufuk Topcu | GPA: 3.94/4.00

Bilkent University June 2021
B.Sc., Electrical and Electronics Engineering | GPA: 3.73/4.00 | High Honors
University entrance exam: 24th of 2 million | Bilkent Comprehensive and Turkish Prime Ministry Scholarships

INDUSTRY EXPERIENCE

Waymo | Software Engineering Intern June - Aug 2026
Planner Team | Mountain View, CA | Python, JAX, SQL

- Research focused on post-training the onboard planner foundation model to unlearn undesirable driving behaviors. Preference learning methods were implemented and benchmarked within Waymo's training infrastructure.
- The evaluation informed a proposed algorithm and data roadmap for Waymo's data flywheel.

Bosch Center for Artificial Intelligence | Research Intern May - Aug 2025
Sunnyvale, CA | Python, PyTorch | First-author paper, arXiv 2026 [1]

- Uniform scenario sampling wastes compute on uninformative driving situations. CL4AD scales curriculum learning to thousands of concurrent self-play agents in GPUDrive through asynchronous episode management, prioritized level replay, and scenario scores for learning progress, safety, and behavioral realism.
- CL4AD reduced wall-clock time to 99% success by 77%, reaching it over a billion steps earlier than uniform sampling. The framework scaled to 80,000 real-world scenarios from the Waymo Open Motion Dataset.

Honda Research Institute USA | Research Intern May - Aug 2024
Ann Arbor, MI | Python, PyTorch, Ray

- A generalist meta-RL teacher adapts to unseen tasks but may perform suboptimally. Gen2Spec distills its knowledge through action advice during student training, accelerating the learning of task-specific RL policies.

Eatron Technologies | Engineering Intern June 2020 - June 2021
Istanbul, Turkey | Python, C++, PyTorch, ROS

- The project targeted trajectory prediction for a Level 2+ advanced driver-assistance system (ADAS). Graph neural networks and self-attention modeled spatiotemporal interactions in traffic scenes to forecast vehicle trajectories.

DOCTORAL RESEARCH

Uncertainty aware offline reinforcement learning
Co-first author | ICLR 2025 [3] | Code

- Offline RL policies can exploit errors in learned dynamics models, yielding optimistic returns that fail to transfer to the real environment. NUNO learns physics-informed dynamics with neural stochastic differential equations, jointly fitting trajectories and separating aleatoric noise from distance-aware epistemic uncertainty.
- The learned uncertainty penalizes synthetic rewards and truncates unreliable rollouts to limit model exploitation. NUNO improved performance by up to 93% over prior offline RL methods on low-quality D4RL and NeoRL MuJoCo datasets.

Safety aware curriculum learning

First author | ICLR 2025 [2] | Code

- Curriculum objectives can be misaligned with constrained RL: tasks selected for reward improvement may undermine safety during training. SCG resolves this misalignment by prioritizing safety before reward in curriculum generation.
- SCG uses constrained optimal transport to expand task coverage subject to safety and performance thresholds. It reduced training-time constraint violations without sacrificing final performance across three constrained RL domains.

Reward synthesis from prior experience and demonstrations

Co-first author | L4DC 2025 [4]

- Sparse rewards hinder exploration beyond a few demonstrated trajectories. The framework combines task-agnostic experience with demonstrations to synthesize dense, dynamics-aware rewards that guide exploration around those trajectories, accelerating optimal-policy learning in robot manipulation even from suboptimal demonstrations.

Robust and structured curricula for reinforcement learning

First-author papers | UAI 2023 [5, 6]

- RACGEN identifies and oversamples rare and risky tasks under heavy-tailed task distributions. It augments an off-the-shelf curriculum, improving expected return as well as returns on challenging tail events. [5] | Code
- Standard curricula overlook the high-level structure of long-horizon tasks. Reward-machine-guided curricula encode subtask dependencies to guide policy learning and task selection, solving tasks where standard curricula failed. [6] | Code

Ongoing research

- Self-play agents can overfit to their own behavior and fail against unfamiliar traffic. Ongoing work investigates curriculum-guided scenario synthesis that varies traffic-behavior composition to improve multi-agent generalization.
- LLM unlearning can damage capabilities that should be retained. I study adaptive curricula for RL post-training that prioritize what to forget while limiting collateral degradation of retained capabilities.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Julia
ML frameworks: PyTorch, JAX, Ray, Hugging Face Transformers, verl, vLLM
Autonomous driving: GPUDrive, PufferDrive
Robotics: MuJoCo, Robosuite, ROS

RESEARCH SERVICE

Reviewer: NeurIPS 2024-26; ICML 2025-26; ICLR 2025-26; AAAI 2026-27; AISTATS 2025-27; TMLR 2026.

Organizer: UT Austin Reinforcement Learning Reading Group | Aug 2025 - Present

SELECTED PUBLICATIONS

[1] Scaling Curriculum Learning For Autonomous Driving
Cevahir Koprulu, David Paz, Feng Tao, et al. arXiv, 2026

[2] Safety-Prioritizing Curricula for Constrained Reinforcement Learning
Cevahir Koprulu, Thiago D. Simão, Nils Jansen, Ufuk Topcu. ICLR, 2025

[3] Neural Stochastic Differential Equations for Uncertainty-Aware Offline RL
Cevahir Koprulu*, Franck Djeumou*, Ufuk Topcu. ICLR, 2025

[4] Dense Dynamics-Aware Reward Synthesis: Integrating Prior Experience with Demonstrations
Cevahir Koprulu*, Po-han Li*, Tianyu Qiu*, et al. L4DC, 2025

[5] Risk-Aware Curriculum Generation for Heavy-Tailed Task Distributions
Cevahir Koprulu, Thiago D. Simão, Nils Jansen, Ufuk Topcu. UAI, 2023

[6] Reward-Machine-Guided, Self-Paced Reinforcement Learning
Cevahir Koprulu, Ufuk Topcu. UAI, 2023

* Equal contribution.